

Exercise 1: Calculate $L(f(t))$ where

$$f(t) = \begin{cases} (t-1)^2 & \text{if } t > 1 \\ 0 & \text{if } 0 \leq t \leq 1 \end{cases}$$

Exercise 2: Find the Laplace Transforms (LT) of the following functions, indicating the values of x for which these LTs exist.

1. $f_1(t) = 5t - 3$

2. $f_2(t) = 3t^4 + 4e^{3t} - 2\sin(5t) + 3\cosh(2t)$

3. $f_3(t) = t^3 e^{-3t}$

4. $f_4(t) = 2e^{3t} \sin(t)$

5. $f_5(t) = \frac{\cos(at) - \cos(bt)}{t}$

6. $f_6(t) = \int_0^t \frac{\cos(au) - \cos(bu)}{u} du$

Exercise 3:

1. Show that $L\left(\frac{\sin^2(t)}{t}\right) = \frac{1}{4} \log\left(\frac{x^2 + 4}{x^2}\right)$

2. Evaluate: $\int_0^{+\infty} \frac{e^{-t} \sin^2(t)}{t} dt$

Exercise 4: Determine the inverse Laplace Transforms of the following functions:

1. $x \mapsto \frac{x^2}{(x+3)^3}$

2. $x \mapsto \frac{1}{x(x+1)^3}$

3. $x \mapsto \frac{x}{(x^2+1)^2}$

4. $x \mapsto \frac{x^3 - 2x^2 + 1}{x^4 + x^2}$

Exercise 5: Solve the differential equation using the Laplace transform:

$$y''(t) + y(t) = \sin(2t) \text{ for } t > 0; \quad y(0) = 2; \quad y'(0) = 0$$